



Investment Society
Imperial College London

Lecture 4

Derivatives I

Securities Education Certificate

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Slide deck inspired by Dr. Euan Doidge and Prof. Anthony Barrett (Department of Chemistry, Imperial College London)

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Background

Some context to introduce you to a concept

Fact

You will need to recall this information in the exam

Model Example

A fully detailed breakdown of an example (question)

Try it yourself

It's your turn to give this problem a go and test your knowledge!

Equation sheet

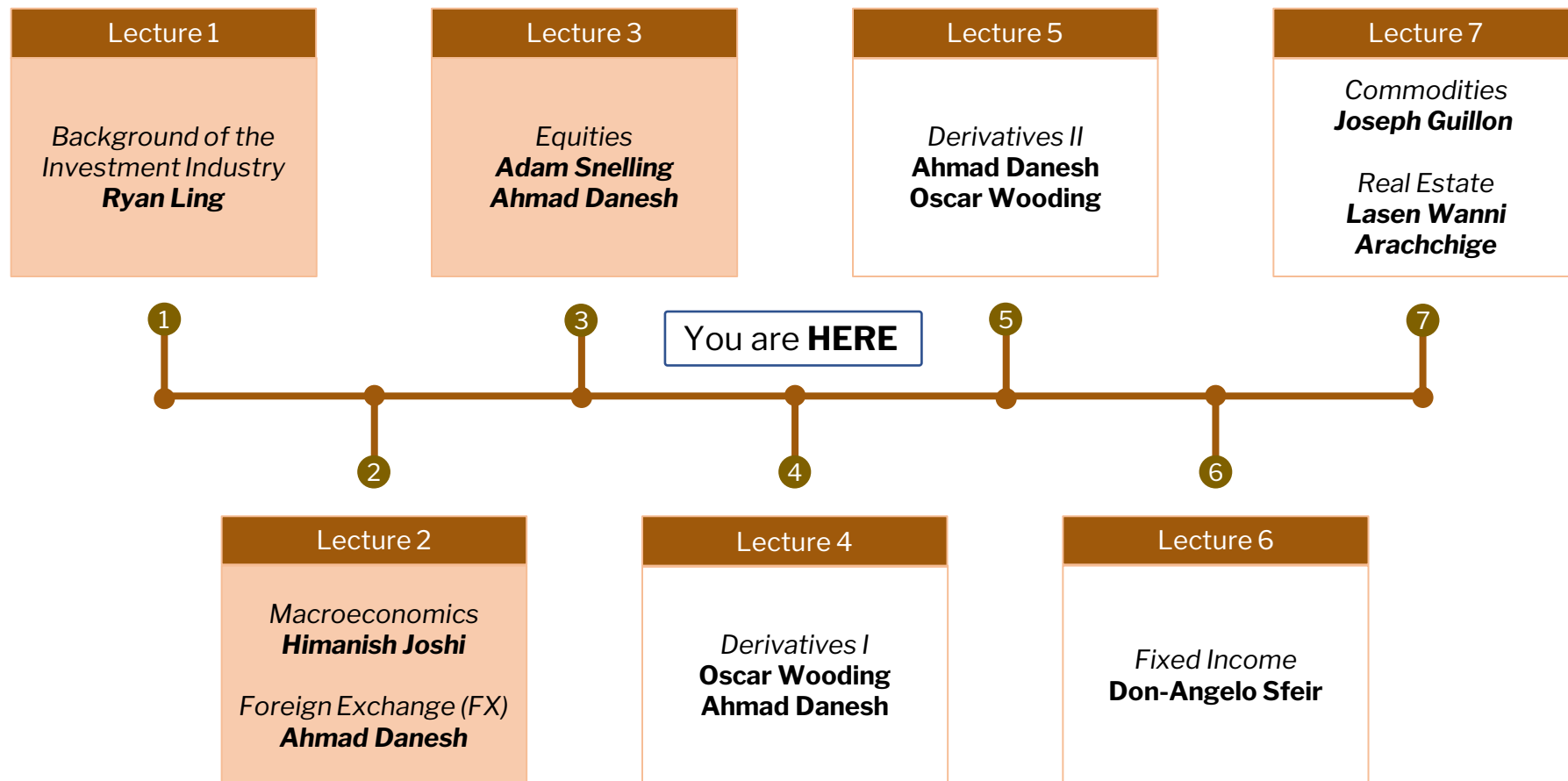
Any equations that you may need for the exam are provided

Taking it further

(non-examinable) stuff to show you where this concept can be applied

The slides are colour-coded to aid your understanding of the course and ensure that you are familiar with the demands of the content provided in each slide.

Lecture Timeline



Let's begin...



Ahmad Danesh
Lead Lecturer
Founder, QT Capital

Lecture 4

A primer on Bonds, Derivatives *Introduction*

Bonds

Is there an alternative way for a company to raise funds?

👉 There are two standard ways for a company to raise funds.

Issue **equity**

Give up ownership of future profits

Issue **debt**

Pay back original amount **WITH** interest

Is there a third way to raise funds that benefits the owners of the *company* the most?

Retain ownership of future profits

Owners set their own interest rate

Your notes

Bonds

*Retaining ownership while setting your own interest rates is a **BOND**.*

👉 Bonds allow borrowers to borrow money on **THEIR** (the borrower's) terms, **NOT** the issuer's.

👉 The borrower sets the interest rate and the size of the money being loaned.

👉 The borrower does **NOT** lose any ownership (equity) in the business.

👉 An interest rate is still paid on the original money loaned.

Government Bonds

LARGE Corporation Bonds

Your notes

Bonds

*Let's look at an example of a UK Government Bond (**'Gilt'**) to show how it works.*

The UK Government wants to raise £1 billion for a new project to pay back in **10 years**.

The Treasury issues bonds of **£1000** to investors → '**par**' / '**face value**'

The *annualised* Interest Rate of the Bond is **3%**.

👉 Bonds are often issued during a **budget deficit** i.e. not enough money to pay for projects.

👉 Investors think the Government is a 'safe' / 'risk-free' bond issuer → Gov. Bonds = 'risk-free return'

Your notes

Bonds

How do we discuss this '**gilt**' in the world of finance and investment?

The UK Government wants to raise £1 billion for a new project to pay back in **10 years**.

Bond Principal = £1000

Coupon Rate = £3%

Maturity Date = 10 years (from issue)

👉 **Bond Principal** → The amount of money a *lender* pays to the *borrower*.

👉 **Coupon Rate** → The interest paid by the *borrower* ('bond issuer') to the *lender* (or the 'bondholder')

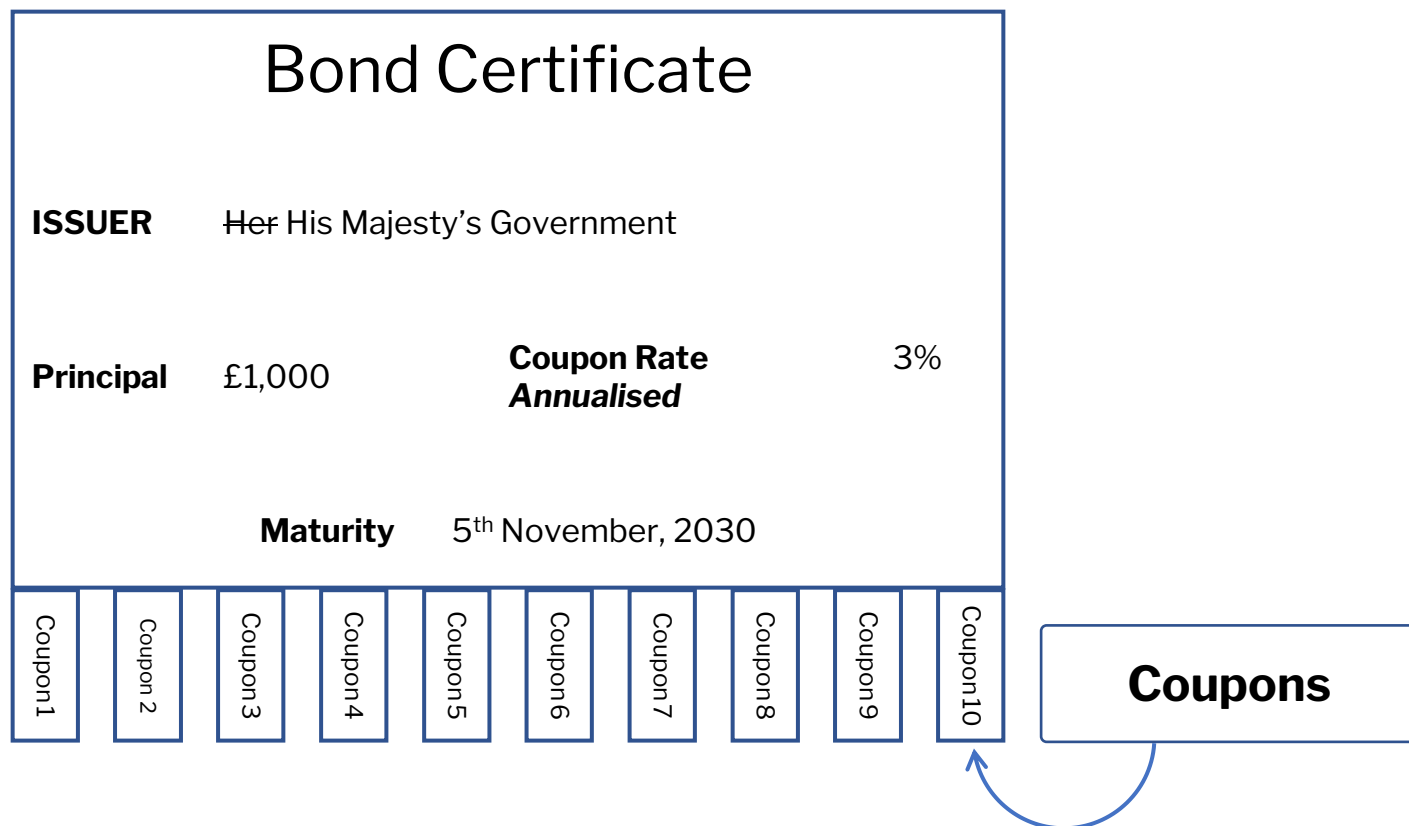
👉 **Maturity** → The date when the bond is paid off completely by the bond issuer.

Your notes

Bond Certificates

How did bonds look in the old days?

👉 Bond issuers published *certificates* for bondholders to track a bond until maturity.

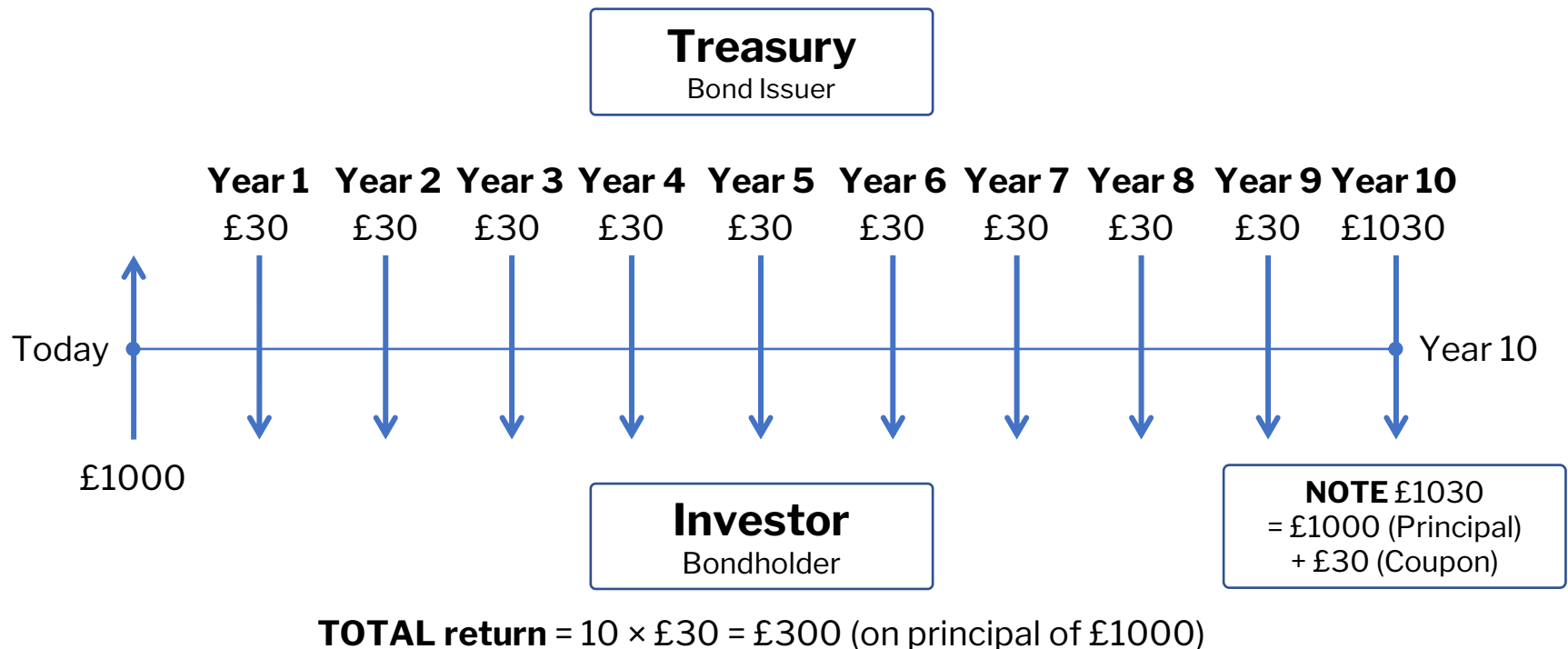


Your notes

Bonds

You can also model bonds in terms of their **cash flow**.

👉 Bonds are just transfers of money between two parties → model the *cash flow*.



Bond Yields

You can also calculate another property of bonds... Bond Yields!

👉 Bond **yields** show the return on a bond for an investor.

$$\text{Bond Yield (\%)} = \frac{\text{Coupon Rate}}{\text{Market Price}} \times 100$$

Why are we dividing by the *Market Price*?

👉 Bonds (c.f. stocks) can be traded on *secondary capital markets* → covers **ALL** bond types (new/old).

👉 If the price of a bond goes down, the bond **yield** goes up → inverse relationship.

Your notes

Coming later...



Don-Angelo Sfeir

Lecturer

Lecture 6

Fixed Income (inc. Bonds)

Definitions

Derivatives are a complex and require lots of definitions. Let's discuss some.

☞ A **risk** → possible *negative* outcomes that can arise from a given event, based on 2 factors.

Likelihood of the event happening

The size of the negative effects

☞ A **counterparty** → the party/syndicate on the other side of the table (in a business deal).

☞ Taking precautionary action against a *risk* → **hedging**.

☞ The risk that you gain from investing into an asset → **exposure**.

Your notes

Managing business risk

Businesses all want to hedge risks, but how do they achieve this?

☞ All investments have **intrinsic risk**.

☞ *Intrinsic* → unavoidable risk to purchasing assets.

☞ Hedging helps to minimise *exposure* to **capital loss**.

☞ A **derivative** → a *contract* that can help investors hedge risk and minimise exposure.

☞ A **contract** binds two or more parties to an agreement with specific terms and conditions.

☞ A **financial derivative** → a contract which bases its value off an *underlying asset*.

Your notes

Underlying Assets

This concept of **UnderLying Asset (UL)** is something you've already seen before...

☞ Car insurance is a derivative contract.

☞ Sport bets are also derivatives contracts.

Insurance Company

Sports Betting Exchange

Car insurance contract

Betting odds contract

Underlying Asset
Car & Driving History

Underlying Asset
Past performance and historical results



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Types of Derivatives in Financial Markets

So, how does this all relate to what you'll see financial markets?

☞ Generally, a derivative contract has 3 parts:

Contract Issuer



Contract



Underlying Assets

☞ This schematic simplifies *most* derivatives.

☞ There are 4 types of *basic* derivatives:

Forwards

Futures

Options

Swaps

Your notes

Let's resume...



Oscar Wooding

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Founder, QT Capital (2021 –)

Lecture 3

Derivatives I

Mechanics of Selling and Shorting

You've heard about selling and shorting an asset, but how do these differ?

☞ Selling is the process of exchanging **an asset that you own** for an agreed payment

☞ Shorting is the process of selling **an asset that you do not own** and buying it back later

Let's consider an example to help understand the mechanism that's used.

☞ I suspect that a Ahmad Corp. is going bankrupt and want to speculate on its share price decreasing.

First, borrow a share of the company (from e.g. a broker/dealer) with interest and the missed dividends.

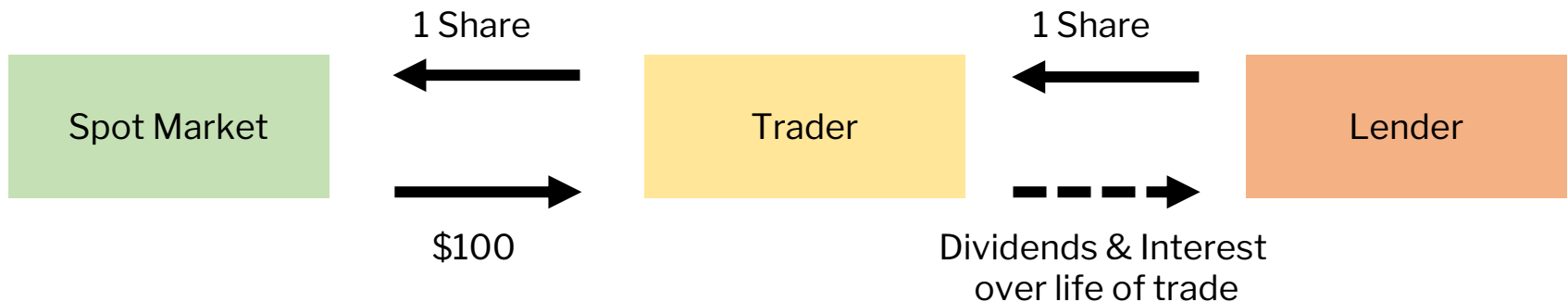
Then, immediately sell the share on the stock market at its spot price of \$100.

Mechanics of Selling and Shorting

First, borrow a share of the company (from e.g. a broker/dealer) with interest and the missed dividends.

Then, immediately sell the share on the stock market at its spot price of \$100.

Trade Opens: $S = \$100, t = 0$



Trade Closes: $S = \$90, t = 1$



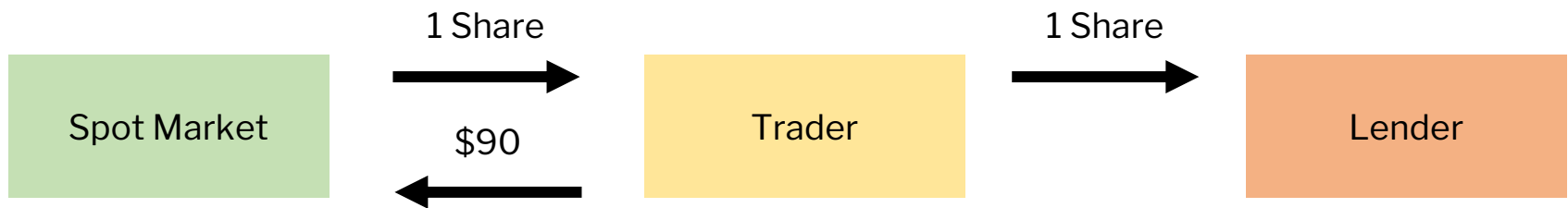
Mechanics of Selling and Shorting

After a year, the price of Ahmad Corp. has decreased to \$90 and we want to take profit.

👉 A short trade is **closed** by buying back the borrowed asset on the spot market.

I can buy 1 share on the spot market and close my initial debt with the lender.

Trade Closes: $S = \$90, t = 1$



Assuming that no dividends or interest are paid, this leaves us a profit of \$10.

Shorting

Shorting is a big part of how Hedge Funds with a higher risk tolerance trade.

☞ Shorting requires no up-front capital and can act as a useful hedge in bearish markets

☞ Unlike a long position, you can lose an unlimited amount – there is no cap on a share's price upward

Short Squeeze

As shorts are on margin, small price movements can force position closure where underlying asset have to be bought back, pushing prices even higher.



Long/Short Strategy

A popular strategy with hedge funds includes using the funds acquired from shorting one asset to create leverage and buy another asset.

Mechanics of Trading FX

FX trades involve buying and selling, shorting can play a part of this process

How can we long a currency pair without having any USD to exchange beforehand?

EUR/USD →

👉 Buy EUR, sell USD on € 10 mn notional

Trade Opens: $S = 1.0000$



Trade Closes: $S = 0.5000$



Mechanics of Trading FX

EUR/USD

👉 Buy EUR & sell USD on €10 mn notional

Trade Opens: $S = 1.0000$



Trade Closes: $S = 0.5000$



Despite the exchange rate only halving, we are left with a **-100%** loss on our position.

Futures and Forwards

Let's start with the hidden time machine invented for grain farmers in 1848!

☞ Both futures and forwards contracts allow investors to do 3 things:

Buy OR sell a fixed amount of an underlying asset

At a fixed price

☞ Common underlying assets:
currencies, commodities and bonds.

☞ The quoted price → how much **must** be
exchanged at expiry.

At a fixed date in the future

☞ This is the **expiry** or **settlement date**.

☞ The date → when the initially agreed price is
exchanged for the fixed amount of asset.

☞ **NOTE** these contracts create an **obligation** between the two parties to exchange assets.

Futures and Forwards

Let's try an example to see how these contracts work.

Imagine you are a farmer and are worried about your cattle herd this year due to weather warnings.

You need to have enough cattle in 2 months time to meet your agreement with supermarkets.

How can we use a forward contract to ensure enough cattle are available?

👉 We enter a **forward** contract on live cattle with an expiry date in 2 months time.

Consider the current price for a cattle 2-month forward contract to be around \$55,000.

The underlier of the contract is 50,000 pounds of live cattle (i.e. 60 animals!) – **You buy 3 contracts.**

Futures and Forwards

Let's try an example to see how these contracts work.

2 months pass and you were right, you are short of 180 cattle and need them urgently.

☞ The expiry on the forward contracts is reached.
You pay a total of \$165,000 for the 180 cattle initially agreed.

☞ If you had not entered the forward contract, you would be forced to buy cattle on the **spot market**.

If the spot market price went up to \$70,000 per 50,000 pounds of cattle, how much did you save?

☞ The forwards contract saved you $(\$70,000 - \$55,000) \times 3 = \underline{\$45,000!}$

☞ The profit from this trade is usually identical to simply buying and holding the underlier

Futures and Forwards – Purpose

Why not just buy the cows and just keep them? Why buy futures and forwards?

☞ You can earn payoffs on the price of an underlying asset without actually owning it.

Without futures or forwards, you would have to e.g. buy and store an actual barrel of oil.

☞ It's very easy to generate large *leverage* as typically **ONLY** profits or losses are exchanged.

☞ The contracts are highly liquid and can be traded very easily electronically with tight margins.

☞ The risk involved with 3rd parties is ~0 as a result of well-developed clearing infrastructure.

☞ US futures are traded 8:30am–3:00pm on weekdays, but usually at least one market is open.

Futures and Forwards – Underliers

These contracts cover almost anything you can spot trade

👉 The most traded instruments on the Chicago Mercantile Exchange (CME) include:

1. S&P 500 E-Mini

2. 10 Year T-Notes

3. Crude Oil

4. 5-Year T-Notes

5. Gold

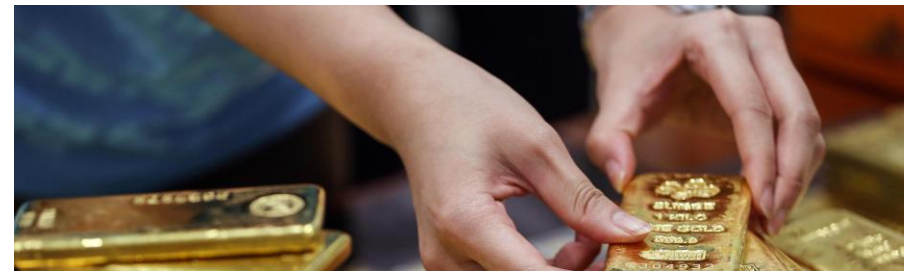
6. EuroFX

7. 30 Year T-Bonds

8. Japanese Yen, JPY

9. 2 Year T-Notes

10. Eurodollars
Dollars held outside the US



Futures and Forwards – Differences

Payoffs for both futures and forwards are identical, but how do the contracts differ?

Futures

Exchange Traded contracts

Traded on standardized exchanges (e.g. CME)

Typically cash settled and market-to-market daily

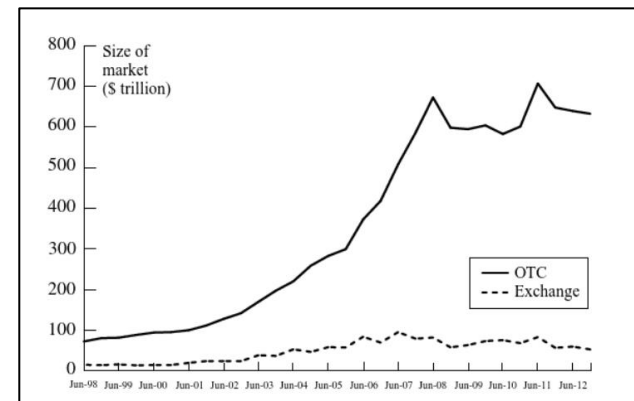


Forwards

Over-the-Counter (OTC) contracts

Custom-made quotes for tailored contracts

Normally for larger deals with complex conditions



OTC vs Exchanges

Payoffs for both futures and forwards are identical, but how do the contracts differ?

Exchange Traded Contracts

Centralised

Fast, efficient and liquid

Closely linked to clearing facilities

Level playing field & trades visible to everyone

Standardized

Over-the-Counter (OTC) Contracts

Networks of broker-dealer relationships

Higher liquidity risk as dealers can withdraw

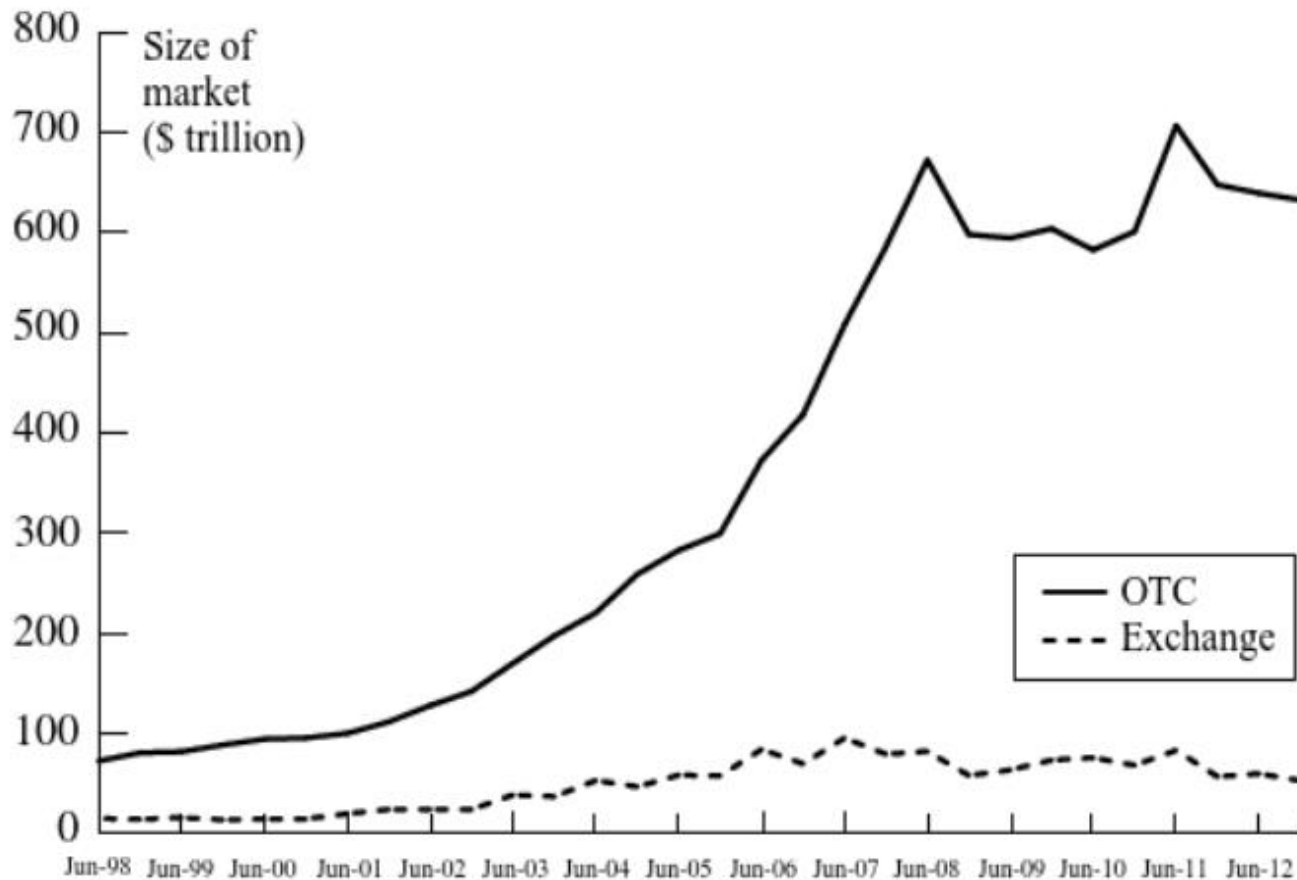
Central Counter Party or bilateral clearing

Dealer's quotes can vary & less transparency

Quotes are tailored to the client

Futures and Forwards – Differences

As **exchange trading** is so **easy** and **quick**, you'd think it was the larger market..



Exchange Mechanics

Exchanges are the main place to acquire derivatives for smaller, simpler trades.

Margin

Amount of funds required in trading account to enter a contract.

Market-to-Market

The value of each position is calculated daily and determines margin used.

👉 A broker will require a certain amount of margin based on the instrument traded

👉 This acts as an assurance that both parties can cover any losses that they incur on a single day.

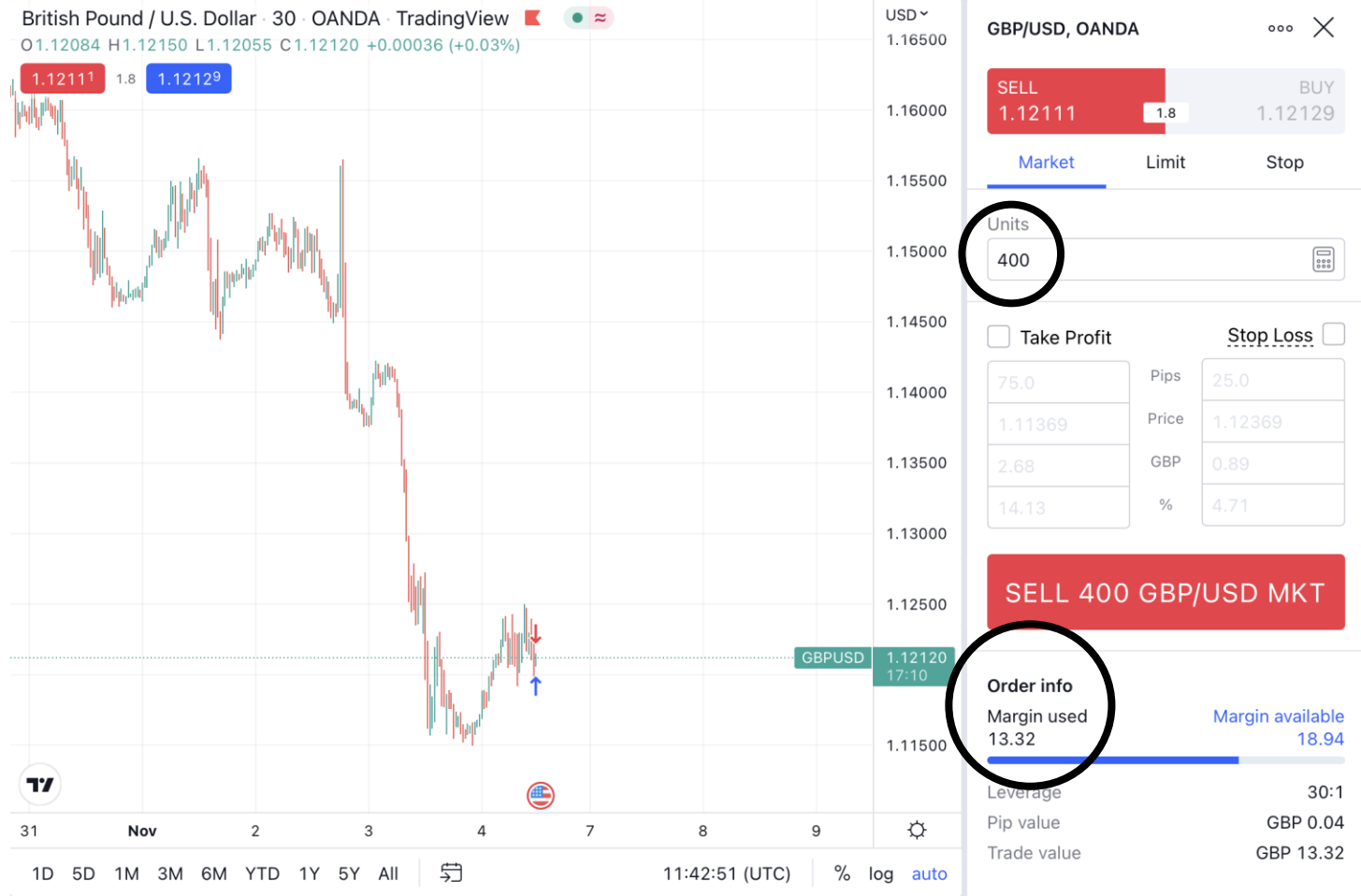
👉 The value of a client's positions are market-to-market and a client's margin account is valued *daily*.

Maintenance Margin

The minimum amount of funds required in a clients margin account to maintain an open position. When breached, a **Margin Call** is triggered.

Exchange Mechanics

Exchanges are the main place to acquire derivatives for smaller, simpler trades.



Exchange Settlement Methods

Exchanges are the main place to acquire derivatives for smaller, simpler trades.

Settlement

Contracts can be cash settled or physically settled.

Cash Settlement

Only **cash** changes hands at the expiry date

Payoff

*Spot price of the contract (at settlement)
– pre-decided futures price.*

Physical Settlement

Purchaser takes ownership of asset

Settlement is done by a Clearing Broker

Less common than cash settlement

Your notes

Cash vs Physical Settlement

Cash Settlement

Very high liquidity

Less time until expiration

Lower costs with often no commission or fees

Simple and does not require offsetting trades

Very popular, especially with commodities

Physical Settlement

Minimal/negligible liquidity

Takes longer
First Notice Day starting delivery

Receiver is responsible to pay for delivery fees

Most contracts are offset to prevent delivery

Involves more risk as goods could be damaged

Exchange Specifications

Standardised

Available contracts cover specific quantity, quality and expiry dates.

CONTRACT UNIT	5,000 bushels
PRICE QUOTATION	U.S. cents per bushel
TRADING HOURS	CME Globex: Sunday – Friday, 7:00 p.m. – 7:45 a.m. CT and Monday – Friday, 8:30 a.m. – 1:20 p.m. CT TAS: Sunday - Friday 7:00 p.m. - 7:45 a.m. and Monday - Friday 8:30 a.m. - 1:15 p.m. CT CME ClearPort: Sunday 5:00 p.m. - Friday 5:45 p.m. CT with no reporting Monday - Thursday from 5:45 p.m. – 6:00 p.m. CT
MINIMUM PRICE FLUCTUATION	1/4 of one cent (0.0025) per bushel = \$12.50 TAS: Zero or +/- 4 ticks in the minimum tick increment of the outright
PRODUCT CODE	CME Globex: ZC CME ClearPort: C Clearing: C TAS: ZCT
LISTED CONTRACTS	9 monthly contracts of Mar, May, Sep and 8 monthly contracts of Jul and Dec listed annually after the termination of trading in the December contract of the current year.
SETTLEMENT METHOD	Deliverable

cmegroup.com/markets.html

Exchange Specifications

Standardised

Available contracts cover specific quantity, quality and expiry dates.



cmegroup.com/markets.html

Over-the-Counter Markets (OTC) Mechanics

CCP

A Central Counter Party

Clears trades between parties to remove counterparty risk through required margin and netting.

Bilateral Clearing

A signed agreement between trading parties covering future transactions and any collateral requirements. Higher counterparty risk than using a CCP.

Closed Forward

Closed forward contracts carry the obligation to be settled at an exact date

Open Forward

Open forwards have a set time window where any part of the contract can be settled. By the end of the window, the full contract must be settled.

Other OTC Contracts

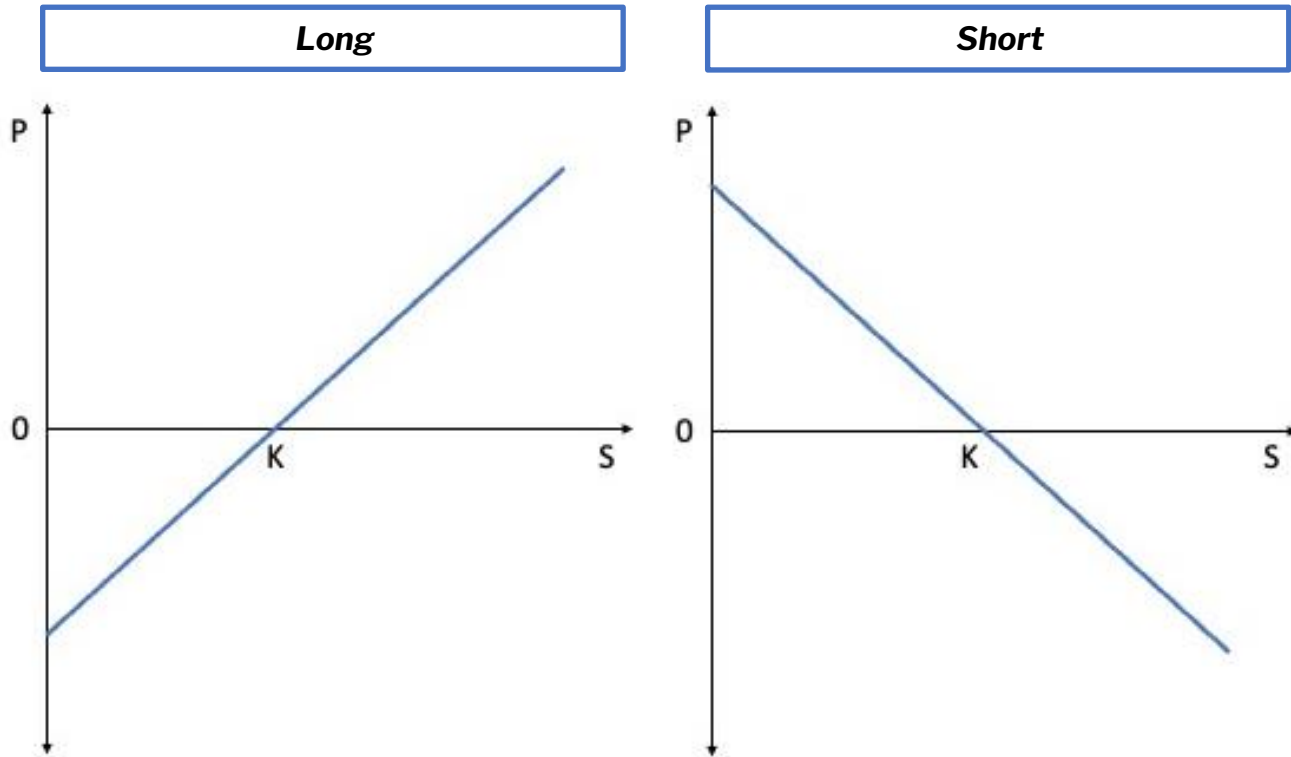
Interest rate swaps, Swaptions, Currency Swaps, CDS, CDO

Poor OTC contract liquidity was a main reasons for the depth of losses in the 2008 Financial Crisis

Illiquid and dysfunctional markets led to an inability for holdings to be valued or sold to limit risk.

Payoffs

The PnL on futures and forwards is identical to purchasing the underlying asset.



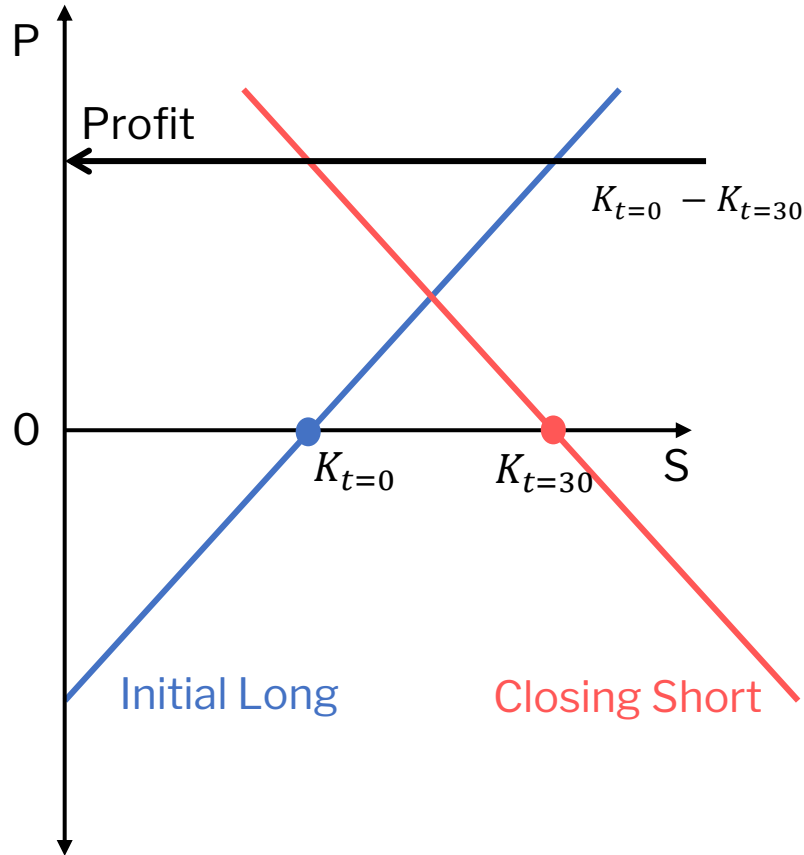
☞ **P** = Profit and Loss (PnL)
on the contract

☞ **K** = quoted price at entry

☞ **S** = Spot price of asset

Payoffs

Now, what if the spot price moves and you want to lock in profit?



You enter a long forward on Gold at K

30 Days pass and the spot has increased

**You can't back out of a contract.
So, how do you take your profit?**

👉 Sell offsetting forward contract at $K_{t=30}$

👉 You are left with a guaranteed profit at expiry per underlier of $K_{t=0} - K_{t=30}$

$$\text{Profit} = \text{red line} + \text{blue line}$$

Futures and Forwards Pricing

How do you determine a fair price for a deal that could be years away?

☞ Futures and Forwards are priced based on several key variables:

Interest Rates

Storage Costs

Convenience Yield

These are all factors which alter the future value of an asset.

☞ Forwards are **only traded on the primary market**

Prices are primarily calculated at contract inception,

☞ Futures instead **only trade on the secondary market**

Prices are made through supply and demand.

☞ Accurate prices should remove any arbitrage opportunities within the *spot* market.

Understanding a Fair Price

What objectives does a fair price have to meet?

Banks

- ☞ Create a contract with no market risk.
- ☞ Use *borrowing* and *lending* to maximal benefit.
- ☞ Avoid use of the bank's capital.

Clients

- ☞ Spot and forwards should be equally attractive in terms of payoff over the contract period
- ☞ Longing and Shorting should be equally attractive.

Forwards are **ALWAYS** priced as the future value of the underlying asset.

Your notes

Interest Conventions

In derivative pricing, both simple and compound interest can be used

Simple Interest

No interest is reinvested

$$F = P \times \left(1 + i \times \frac{D}{B}\right)$$

F = Future Value
P = Principal
i = Interest Rate (per year)
D = Interest Period (days)
B = Day-Count Basis (normally 360, exception of 365 for UK)

Compound Interest

Reinvestment after each interest period

$$F = P \times (1 + i)^t$$

F = Future Value
P = Principal
i = Interest Rate (per year)
t = time (years)

When is each convention used?

For interest periods below a year, we use **simple interest**. Otherwise, **compound interest** is used.

Pricing with Interest Rate Exposure

Let's start with an example from the client's perspective.

As a client, we want to buy **\$10 million principal T notes** (with maturity in 3 years), **in 1 year**

Spot price of 5% Coupon 3 year T Notes is 100 (par).

We assume that the bond is held to maturity and so the price does not change

We also assume a risk-free rate of 2% - risk free interest from the government

We require **delivery 1 year after the spot date.**

What price makes buying the bond now and entering the forward equally attractive?

Pricing Bond Forwards

Let's consider the two options that the client has.

Buy the spot 4yr T Note now

Pay T Note forward rate (X) in 1 year

Today...

Purchase of the bond

- \$100

Invest spare funds
at RFR

- \$100

In 1 year...

5% Bond Coupon

+ \$5

Invest spare funds
at RFR

+ \$102

Purchase the bond at the
forward rate (X)

- \$X

 **The client is left with a bond worth \$100 and \$5.**

 **The client receives the same bond and \$102 - \$X**
X is just a number.

Pricing Bond Forwards

Let's consider the two options that the client has.

Buy the spot T Note now

Pay T Note forward rate in 1 year

In 1 year...

☞ ***The client is left with a 3yr T Note and \$5***

☞ ***The client receives the same bond and \$102 - \$X***

☞ To make both scenarios equivalent, the forward rate (X) must be \$97

This is the same as the future value of the bond minus the convenience yield.

$$F = S \times ((1 + r)^t - (1 + d)^t + 1)$$

F = forward price, S = spot, r = risk free rate, d = convenience yield (gain from asset), t = time (years)

Pricing Forwards (settles in over a year)

Let's look at one way which you could derive this equation

We set both possible paths to owning the asset **in t year** to be equal in value.

$$S \times ((1 + d)^t - 1) = S \times (1 + r)^t - F$$

2. Return from purchasing spot

1. Returns from Forward



$$F = S \times ((1 + r)^t - (1 + d)^t + 1)$$

F = forward price, S = spot, r = risk free rate (i.e. 0.02), d = convenience yield (i.e. 0.05), t = time (years)

Convenience yield is the income generated by the underlying asset per annum.

This formula could also be adapted to include **storage costs** (for e.g. oil) by using this technique.

Pricing Forwards (settles in under a year)

Let's look at one way which you could derive this equation

We set both possible paths to owning the asset **in D days** to be equal in value.

$$S \times (d \times \frac{D}{B}) = S \times (1 + r \times \frac{D}{B}) - F$$

2. Return from purchasing spot

1. Returns from Forward



$$F = S \times (1 + (r - d) \times \frac{D}{B})$$

F = forward price, S = spot, r = risk free rate (i.e. 0.02),
 d = convenience yield (i.e. 0.05), D = day count (years), B = day count basis

In reality, the day on which the dividend is paid would impact the price, but this is a good estimate.